

Yulong Zhao

Statistics PhD Candidate | Statistical Machine Learning & Sequential Modeling

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Summary

Statistics PhD candidate building interpretable models for user behavior sequences: how each action's effect changes over time, which latent skills and states drive behavior, and when to raise early warnings with calibrated false alarms.

Education

University of Arizona — PhD, Statistics	Expected May 2028
University of Wisconsin–Madison — MS, Statistics and Data Science	2023
University of Science and Technology of China — BS, Mathematics and Applied Mathematics	2022

Technical Skills

Programming: Python, R.

Modeling: latent-variable models, EM, regularized GLMs, sequence models (GRU, TCN, Transformer), time-series forecasting.

Tools: NumPy, pandas, scikit-learn, TensorFlow/Keras, Numba, Git, Slurm/HPC.

Evaluation: cross-validation, bootstrap, walk-forward validation, threshold calibration, multiple-testing correction.

Research Experience

University of Arizona — PhD Researcher 2023 – Present

Advisors: Prof. Xueying Tang, Prof. Hao Helen Zhang | NSF #2310664, #1929348

Time-Varying Action Effects in User Behavior Sequences

- Developed an **interpretable model** that estimates how each action's effect on task success changes with *when* it occurs, classifying every action as having no effect, a constant effect, or a time-varying effect.
- Applied it to 19,089 events from 2,668 users in an educational problem-solving assessment (OECD PISA 2012), revealing actions whose effect **reverses over task time**, a pattern masked by aggregate action counts.
- Also improved prediction over a count-feature logistic baseline: **50% lower log loss** ($0.237 \rightarrow 0.118$) and accuracy from 93.5% to 96.1% in 10-fold cross-validation, winning every fold.

Early Failure Detection from Partial User Sequences

- Built an **online early-warning framework** that updates each user's failure risk after every action in educational problem-solving tasks, enabling intervention before task completion.
- Calibrated alert thresholds** to target a user-level false-alarm rate of $\leq 5\%$.
- Benchmarked GRU, temporal CNN, Transformer, and leakage-safe logistic baselines; the best temporal CNN detected **98.9%** of failing users on average across three tasks (97.6% on the hardest) at precision ≥ 0.98 , with false alarms at most 4% on every task, vs. 85.6% for GRU.

Learning Latent Skills and Behavioral States

- Developed a **structured latent-variable model** that jointly estimates persistent user skills and evolving behavioral states from multistep interaction sequences.
- Learned sparse mappings linking latent skills to actions, interface choices, and state transitions, without prespecified skill-behavior assignments or manually labeled states.
- On 1,340 users of an email-and-spreadsheet task, **recovered interpretable structure** without labels: distinct skills for search, sort, and email actions, skills with opposing effects on related actions, and state transitions consistent with the task logic.

Time-Series Forecasting & Regime-Shift Detection

- Built a **PCA + RBF-SVR** forecasting pipeline with leakage-free walk-forward evaluation, reducing MSE by **38%** on average versus a historical-mean baseline and outperforming it in all 18 settings ($3 \text{ outcomes} \times 3 \text{ fossil records} \times 2 \text{ preprocessing choices}$).
- Evaluated environmental regime shifts using **recurrence plots**, multiple-testing correction, and preprocessing sensitivity analysis, distinguishing robust findings from preprocessing-dependent results.

Selected Research Outputs

Zhao, Y. & Tang, X. "Scalar-on-Process Model for Educational Assessment." *Manuscript ready for submission*.

Refereed oral presentation, Symposium on Data Science & Statistics (SDSS), 2026.